

1. Capture an images using a webcam or dataset. Apply image preprocessing (grayscale, histogram equalization).
2. Convert a color image to grayscale. Display the histogram of the grayscale image.
3. **Average Filtering:**
 - Apply an **average (mean) filter** with different kernel sizes (3×3 , 5×5 , 7×7).
 - Observe and compare the smoothing effect on noisy images.
 - Plot the original and filtered images side by side.
4. **Median Filtering:**
 - a. Apply a **median filter** on the same noisy image.
 - b. Compare its performance with the average filter, especially for salt-and-pepper noise.
5. **Unsharp Masking:**
 - Apply **unsharp masking** to enhance edges in a grayscale image.
 - Experiment with different mask weights (e.g., 0.5, 1.0, 1.5).
6. **Laplacian/Edge Enhancement:**
 - Apply a **Laplacian filter** to highlight edges.
7. **Gradient-Based Edge Detection:**
 - Implement **Sobel** and **Prewitt** operators to detect edges.
8. **Fourier Transform of Images:**

Compute the **2D Fourier Transform** of a grayscale image.

Display the magnitude spectrum of the image.
9. Apply smoothing and sharpening in both spatial and frequency domains. Compare the visual quality and computation time.
10. Load a grayscale image. Apply **global thresholding** to convert it into a binary image. Experiment with **Otsu's automatic thresholding** and compare results
11. Load an image
 - Apply **dilation** on a binary image and observe how object boundaries grow.
 - Apply **erosion** and observe how object boundaries shrink.
 - Test different structuring elements (square, circle) and sizes.
12. Implement **opening** (erosion followed by dilation) and **closing** (dilation followed by erosion). Apply to a noisy binary image and compare results for noise removal and object preservation.
13. **Labeling Objects in Binary Images:**
 - Perform **connected component analysis** on a binary image.
 - Assign unique labels/colors to each connected component.
 - Count the total number of objects and compute their area.

14. Implement **region growing** or **split-and-merge segmentation** on a grayscale image.
 - Display segmented regions with different colors.
 - Discuss how seed selection affects the segmentation result.
15. Apply **watershed algorithm** to segment overlapping objects.
 - Use gradient magnitude images to detect object boundaries.
 - Display the segmented regions and contours on the original image.
16. Detect objects in a binary or segmented image. Draw **bounding boxes** or **minimum enclosing rectangles** around detected objects. Count the number of detected objects.
17. Read a sample MPEG video.
 - Extract frames and display them sequentially.
 - Convert frames to grayscale or binary images for analysis.
18. Detect moving objects in a video using **frame differencing** or **background subtraction**.
 - Apply morphological operations to clean the segmented objects.
 - Draw bounding boxes around moving objects in real-time.
19. Load a grayscale image.
 - Compute the **LBP histogram** for the image.
 - Visualize the LBP image and histogram.
20. Compute **GLCM features** (contrast, correlation, energy, homogeneity) from a grayscale image.
 - Compare the texture descriptors for different regions of the image.
21. Detect **SIFT keypoints and descriptors** in two images.
 - Match features between images using **Brute-Force Matcher** or **FLANN**.
 - Visualize matched keypoints with lines connecting correspondences.
22. Detect **SURF keypoints and descriptors** for the same images.
23. Implement a **Bayes classifier** for a 2-class synthetic Gaussian dataset.
 - Compute class probabilities and classify test points to minimize misclassification rate.
 - Plot decision boundary.
24. Load a dataset (e.g., Wine or Breast Cancer dataset).
 - Perform feature selection → apply LDA → train a Bayes classifier considering loss matrix.
 - Evaluate performance using **accuracy, confusion matrix, and expected loss**.
25. Load a pre-trained **YOLOv5 or YOLOv8 model**.

- Apply YOLO to detect objects in images or a short video clip.
- Draw bounding boxes with class labels and confidence scores.
- Compute **precision, recall, and mAP (mean Average Precision)** for detected objects.

26. Content-Based Image Retrieval (CBIR)

Feature Extraction for CBIR:

- Extract features from a set of images using **color histograms, LBP, or HOG**.
- Store feature vectors for each image in a small dataset.

Image Retrieval Based on Similarity:

- Given a query image, compute **Euclidean or Bhattacharya distance** to all dataset images.
- Retrieve and display the **top 3 most similar images**.
- Discuss the effect of using different feature descriptors on retrieval accuracy.